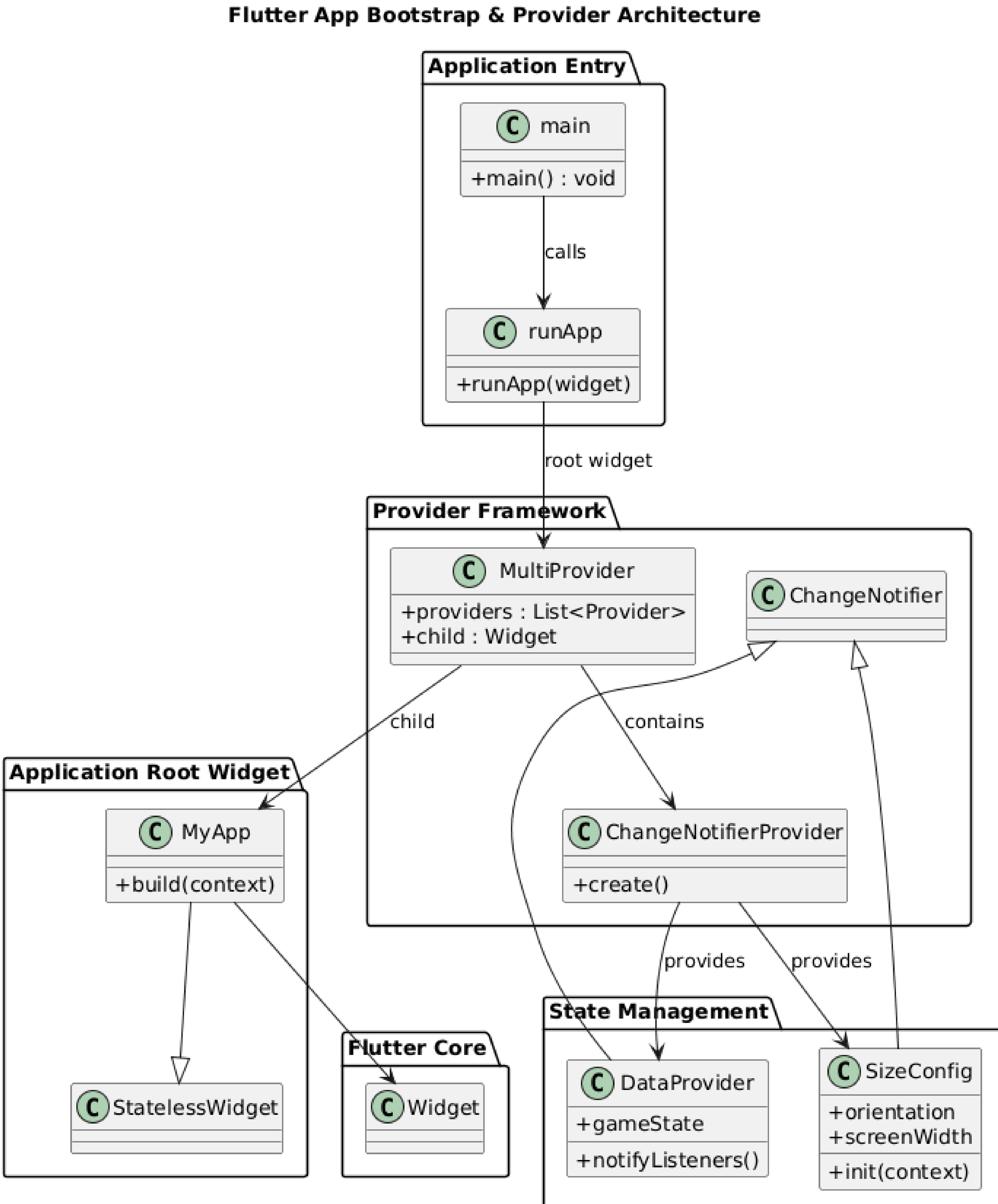
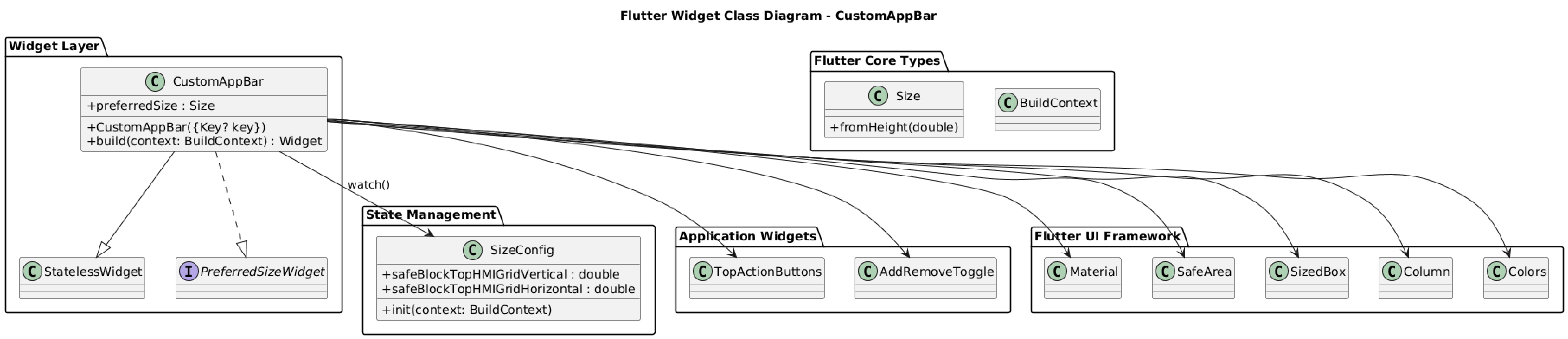
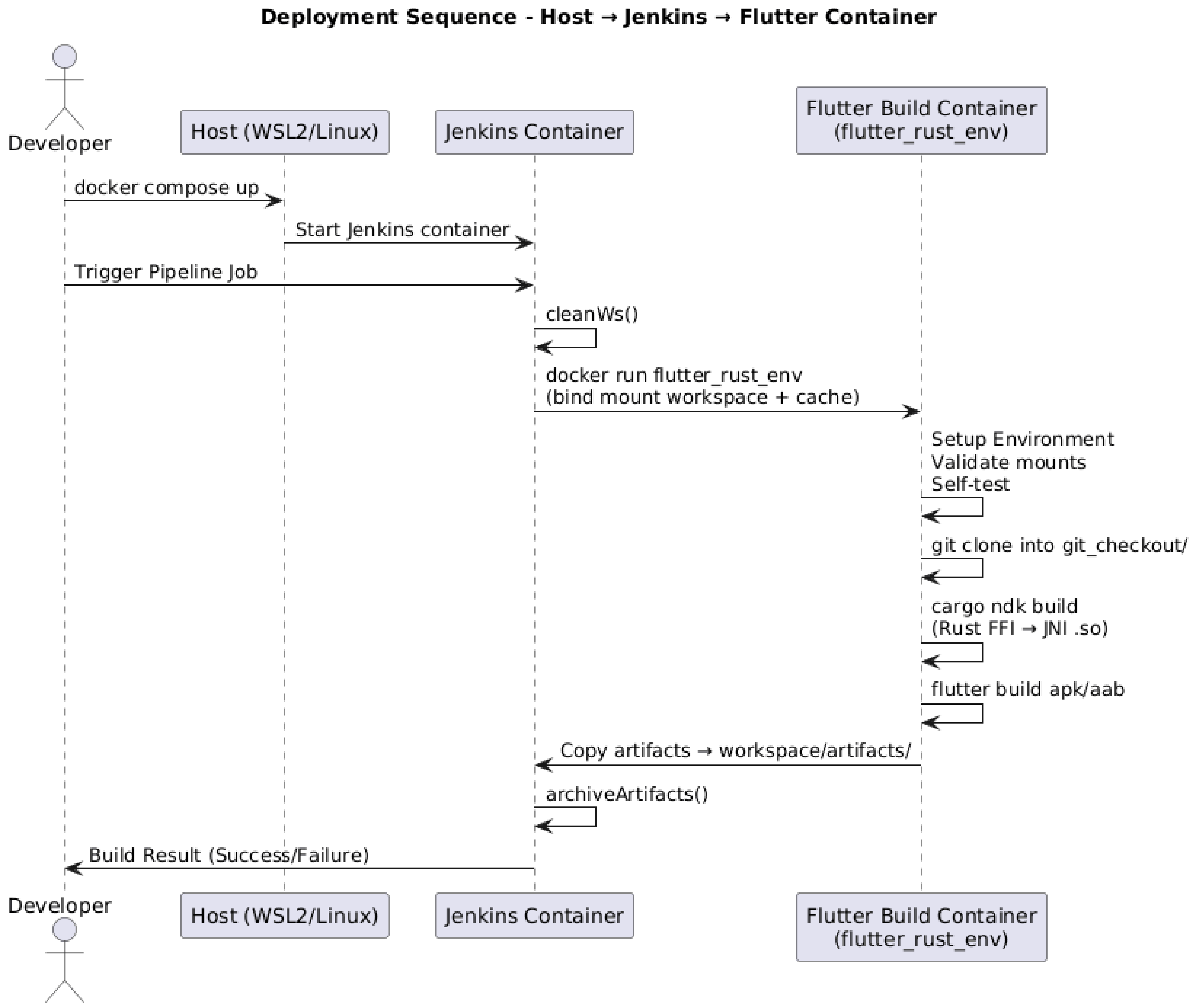
Flutter Widget Class Diagram - AddRemoveToggle


Flutter App Bootstrap & Provider Architecture
Application Entry
main
+main() : void
calls
runApp
+runApp(widget)
root widget
Provider Framework
MultiProvider
+providers : List<Provider>
+child : Widget
ChangeNotifier
child
contains
ChangeNotifierProvider
+create()
Application Root Widget
MyApp
+build(context)
StatelessWidget
Flutter Core
Widget
provides
provides
State Management
DataProvider
+gameState
+notifyListeners()
SizeConfig
+orientation
+screenWidth
+init(context)



Flutter Widget Class Diagram - CustomAppBar
Widget Layer
CustomAppBar
+preferredSize : Size
+CustomAppBar({Key? key})
+build(context: BuildContext) : Widget
StatelessWidget
PreferredSizeWidget
watch()
State Management
SizeConfig
+safeBlockTopHMIGridVertical : double
+safeBlockTopHMIGridHorizontal : double
+init(context: BuildContext)
Flutter Core Types
Size
+fromHeight(double)
BuildContext
Application Widgets
TopActionButtons
AddRemoveToggle
Flutter UI Framework
Material
SafeArea
SizedBox
Column
Colors

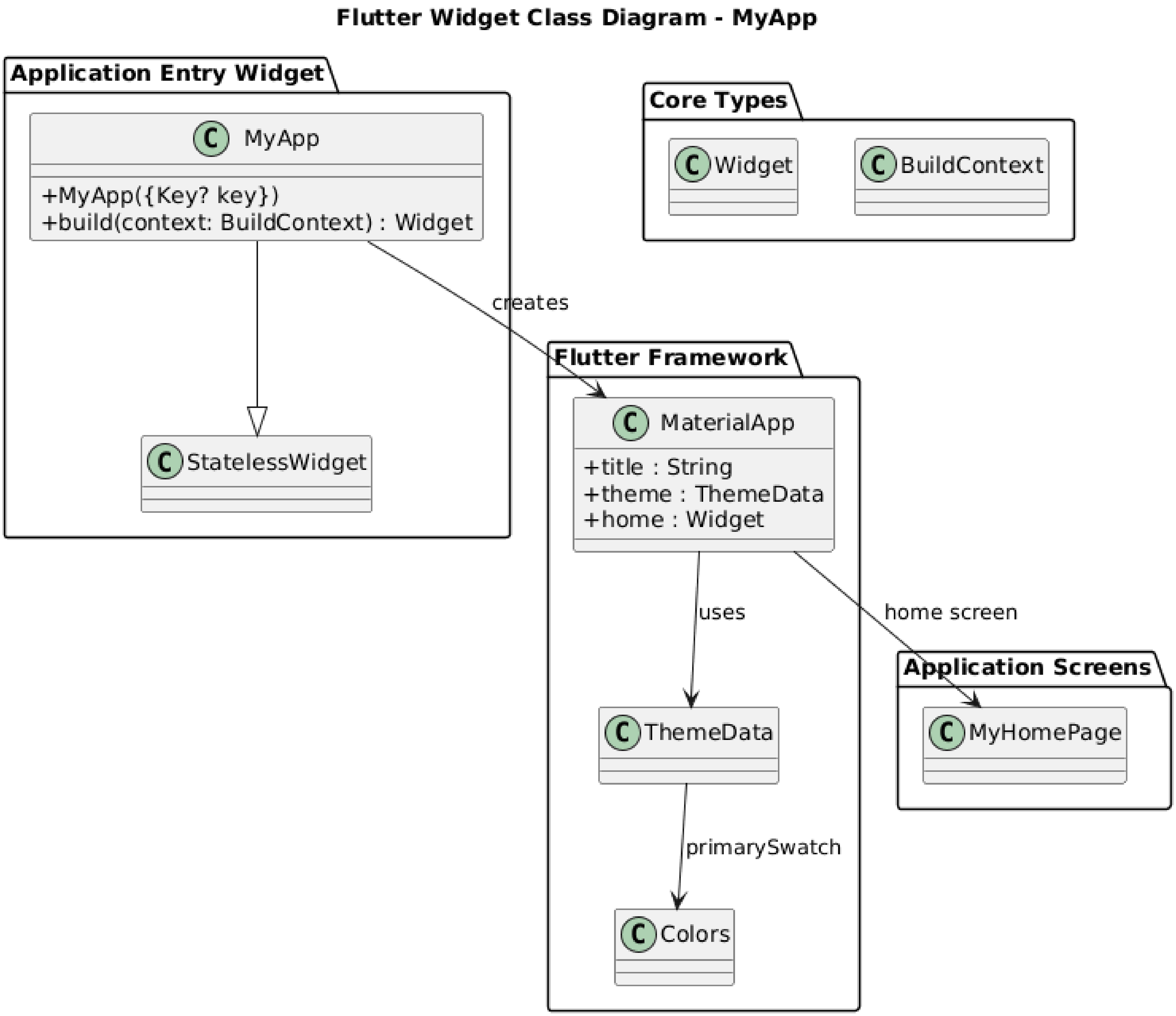

Deployment Sequence - Host → Jenkins → Flutter Container
Developer
Host (WSL2/Linux)
Jenkins Container
Flutter Build Container
(flutter_rust_env)
docker compose up
Start Jenkins container
Trigger Pipeline Job
cleanWs()
docker run flutter_rust_env
(bind mount workspace + cache)
Setup Environment
Validate mounts
Self-test
git clone into git_checkout/
cargo ndk build
(Rust FFI → JNI .so)
flutter build apk/aab
Copy artifacts → workspace/artifacts/
archiveArtifacts()
Build Result (Success/Failure)


Immutable Toolchain Versions
Java : 17
Flutter : 3.35.7
Rust : 1.91.1
Android SDK Tools : 13114758
Android Compile : 36
Android BuildTools: 35.0.0
NDK : 28.2.13676358
CMake : 3.22.1
Base Image : ubuntu 22.04
Docker Multi-Stage Build
env
Defines:
- All ENV variables
- Tool versions
- Paths
base
Adds:
- System packages
- OpenJDK 17
- 32-bit libraries
- build-essential
android
Installs:
- Android cmdline tools
- platform-tools
- SDK 36
- Build Tools 35.0.0
- NDK 28.2.13676358
- CMake 3.22.1
rust
Installs:
- Rust 1.91.1
- Android targets
- cargo-ndk
chrome
Installs:
- google-chrome-stable
- Headless flags
flutter
Installs:
- Flutter 3.35.7
- flutter precache --android
- Locks SDK read-only
final
Assembles:
- Android SDK (read-only)
- Flutter SDK (read-only)
- Rust toolchain
- Chrome
- Runs as UID 2000
Security Model
Persistent Host Cache
(Mounted Volume)
/home/jenkins_host_cache
.pub-cache
.gradle
.cargo
Writable
UID 2000
Speeds up builds
No SDK content
Immutable Toolchain Layer
(Inside Image)
/opt/flutter
/opt/android/sdk
/opt/rust
/usr/lib/jvm
Installed at build time
Owned by root
Read-only in CI
Fully reproducible
Ephemeral Workspace
(Mounted)
CONTAINER_WORKSPACE
git checkout
build outputs
android/build
.dart_tool
Writable
Clean per build
No toolchain
Reproducibility Model
Pinned Versions
ENV-based version locking:
- FLUTTER_VERSION
- RUST_VERSION
- BUILD_TOOLS
- NDK_MAIN
- COMPILE_SDK
Immutable SDK Lock
chmod -R a=rX
SDK never modified at runtime
Non-root Runtime
USER 2000
Prevents SDK mutation


Jenkins Sudoku CI - Docker Compose Deployment
Host Machine
Linux
Host Resources
Docker Socket
/var/run/docker.sock
Jenkins Workspace
/home/mirko/jenkins_host_workspace
Docker Engine
Container
jenkins_sudoku_container
Docker Group Access
GID:1001
Volumes
/home/mirko/jenkins_host_workspace -> /var/jenkins_home
/var/run/docker.sock -> /var/run/docker.sock
Exposed Ports
8080 -> Jenkins Web UI
50000 -> Jenkins Agent Port
Jenkins Server
Environment Variables
TZ = Europe/Paris
JAVA_OPTS=-Djenkins.install.runSetupWizard=false
Image
jenkins_sudoku_image:2.528.3
User Runtime
UID:2000
GID:2000
Docker Compose
Service: jenkins
Docker Network
jenkins_net (bridge)
Developer / DevOps
docker compose up
build image
Docker Build System
Dockerfile
(Current Directory)
Build Context
.

Jenkins CI Pipeline - Flutter + Rust + Android
Workspace Sanity Clean
cleanWs()
Ensures fully fresh ephemeral workspace.
Define Project Paths
Sets:
FLUTTER_CONTAINER_WORKSPACE = $WORKSPACE
FLUTTER_CONTAINER_CACHE = /jenkins_container_cache
REPO_CHECKOUT_DIR = workspace/git_checkout
Separates workspace vs persistent cache.
Setup Environment
Creates:
- .gradle
- .pub-cache
- .home
- artifacts/
Validates:
- Cache writable
- SDK read-only
CI Self-Test
Fail-fast validation:
✔ Required ENV variables exist
✔ Workspace & cache mounted
✔ Toolchains available
✔ Flutter SDK NOT writable
Source Control
Checkout in Container
git clone → git_checkout/
git reset to branch
Gradle wrapper executable
Validate Repo Structure
Requires:
- pubspec.yaml
- android/
- scripts/
Verify Rust Source
Ensures rust/rust_lib exists.
Static Analysis (Optional)
RUN_FAST_STATIC?
yes
Flutter + Rust FAST
flutter analyze
dart format
cargo check
RUN_HEAVY_STATIC?
yes
Android + Rust HEAVY
./gradlew lint
cargo clippy
cargo audit
Cleaning Strategy
Light Clean
Removes:
- build/
- android/build/
- .dart_tool
Keeps persistent caches intact.
DEEP_CLEAN?
yes
Deep Clean
Also removes:
- Gradle module caches
- Pub hosted/git
- Rust artifacts
- JNI libs
Build Graph
Build Rust (FFI)
cargo ndk build --release
Produces JNI .so libraries
into android/app/src/main/jniLibs
RUN_MATERIALIZATION_STAGE?
yes
Flutter pub get
flutter build apk --debug
Validates Flutter ↔ Gradle
integration and engine artifacts.
Release Build
yes
BUILD_MODE == release?
debug
Inject Android Signing
Uses Jenkins credentials:
- KEYSTORE_FILE
- KEYSTORE_PASSWORD
- KEY_PASSWORD
- KEY_ALIAS
Generates key.properties
Build Release APK
Build Release AAB
flutter build apk --release
flutter build appbundle --release
Copies output to artifacts/
Build Debug APK
Validation & Extras
RUN_INTEGRATION_TESTS?
yes
Run Integration Tests
RUN_ABB_RELEASE_TEST?
yes
Validate AAB
Uses bundletool:
✔ Validate structure
✔ Verify signature
✔ Dump versionCode/versionName
RUN_PLANTUML_DOCU_BUILD?
yes
Generate Diagrams
Archive Artifacts
archiveArtifacts artifacts/**
Stored under:
Jenkins build archive
Post Section
✔ Remove key.properties
✔ cleanWs(deleteDirs=true)
✔ Print build result

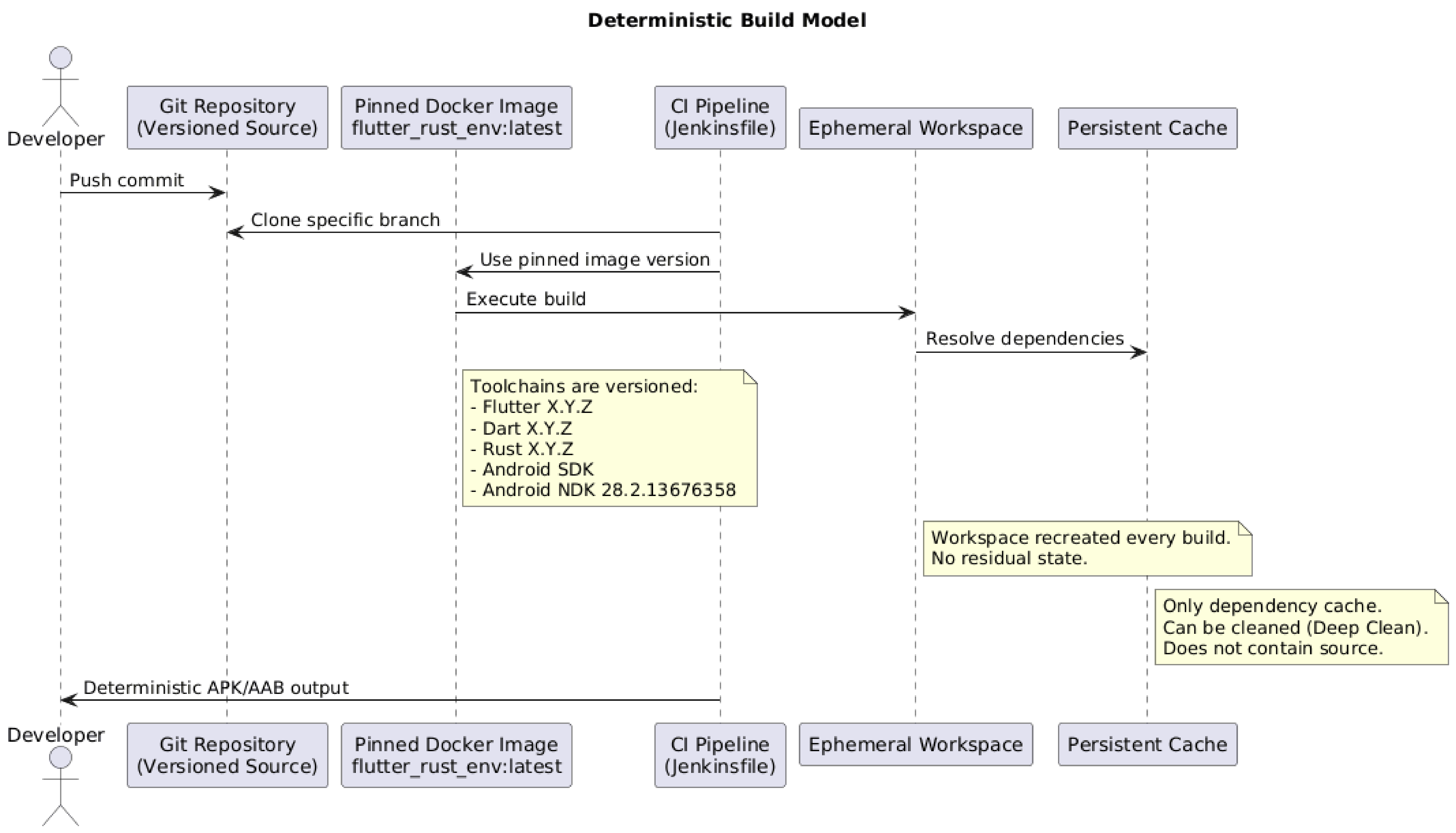
Jenkins Sudoku CI - Docker Image Build
Developer / CI Pipeline
Docker Build Process
Build Command
docker build -t jenkins_sudoku_image:2.528.3 .
docker build --no-cache -t jenkins_sudoku_image:2.528.3 .
Dockerfile
(Current Directory)
Base Image Layer
jenkins/jenkins
2.528.3-lts-jdk21
System Configuration Layer
User Configuration
USER root
usermod -u 2000 jenkins
groupmod -g 2000 jenkins
Git Configuration
git config --system
safe.directory /opt/flutter
Jenkins Setup Configuration
JAVA_OPTS=
-Djenkins.install.runSetupWizard=false
Package Installation Layer
APT Dependencies
docker-cli
ca-certificates
curl
git
Jenkins Plugin Layer
Jenkins Plugins
git
workflow-aggregator
github
github-branch-source
pipeline-stage-view
docker-workflow
ws-cleanup
Diagram Tooling Layer
Documentation Tools
plantuml
graphviz
Docker Integration Layer
Docker Group Configuration
groupadd -g 1001 docker
usermod -aG docker jenkins
Final Runtime Image
jenkins_sudoku_image:2.528.3
Runtime User
jenkins (UID:2000)

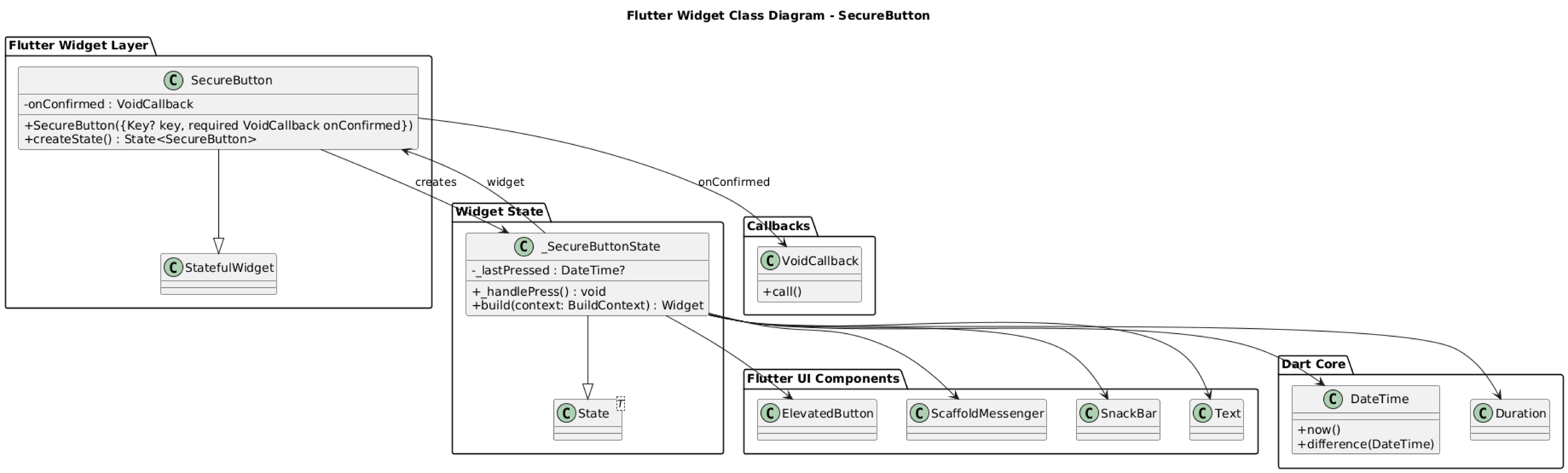

Flutter Widget Class Diagram - MyApp
Application Entry Widget
MyApp
+MyApp({Key? key})
+build(context: BuildContext) : Widget
StatelessWidget
creates
Core Types
Widget
BuildContext
Flutter Framework
MaterialApp
+title : String
+theme : ThemeData
+home : Widget
uses
home screen
ThemeData
primarySwatch
Colors
Application Screens
MyHomePage


Flutter Widget Class Diagram - MyHomePage
Widget Layer
MyHomePage
+MyHomePage({Key? key})
+build(context: BuildContext) : Widget
-_buildPortraitLayout(sizeConfig: SizeConfig) : Widget
-_buildLandscapeLayout(sizeConfig: SizeConfig) : Widget
StatelessWidget
watch()
State Management
SizeConfig
+orientation : Orientation
+screenWidth : double
+safeBlockTopHMIGridVertical : double
+safeBlockBottomHMIGridVertical : double
+safeBlockTopHMIGridHorizontal : double
+init(context: BuildContext)
Application Widgets
SudokuGrid
CustomAppBar
ToggleButtonsSample
Flutter UI Framework
Scaffold
AnimatedSwitcher
SafeArea
Column
Row
LayoutBuilder
Expanded
Center
Container
SizedBox
Material
layout decision
Flutter Core Types
BuildContext
Orientation

Flutter Widget Class Diagram - NumberButtons

Flutter Widget Class Diagram - PatternButtons

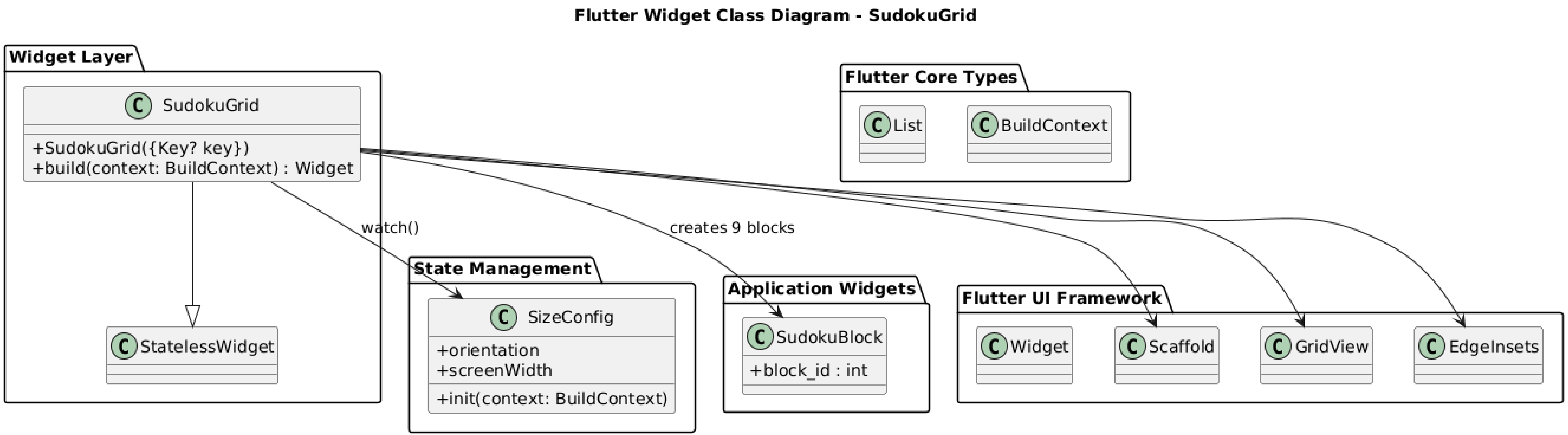

Deterministic Build Model
Developer
Git Repository
(Versioned Source)
Pinned Docker Image
flutter_rust_env:latest
CI Pipeline
(Jenkinsfile)
Ephemeral Workspace
Persistent Cache
Push commit
Clone specific branch
Use pinned image version
Execute build
Resolve dependencies
Toolchains are versioned:
- Flutter X.Y.Z
- Dart X.Y.Z
- Rust X.Y.Z
- Android SDK
- Android NDK 28.2.13676358
Workspace recreated every build.
No residual state.
Only dependency cache.
Can be cleaned (Deep Clean).
Does not contain source.
Deterministic APK/AAB output


Flutter Widget Class Diagram - SecureButton
Flutter Widget Layer
SecureButton
-onConfirmed : VoidCallback
+SecureButton({Key? key, required VoidCallback onConfirmed})
+createState() : State<SecureButton>
StatefulWidget
creates
widget
onConfirmed
Widget State
_SecureButtonState
-_lastPressed : DateTime?
+_handlePress() : void
+build(context: BuildContext) : Widget
State
T
Callbacks
VoidCallback
+call()
Flutter UI Components
ElevatedButton
ScaffoldMessenger
SnackBar
Text
Dart Core
DateTime
+now()
+difference(DateTime)
Duration

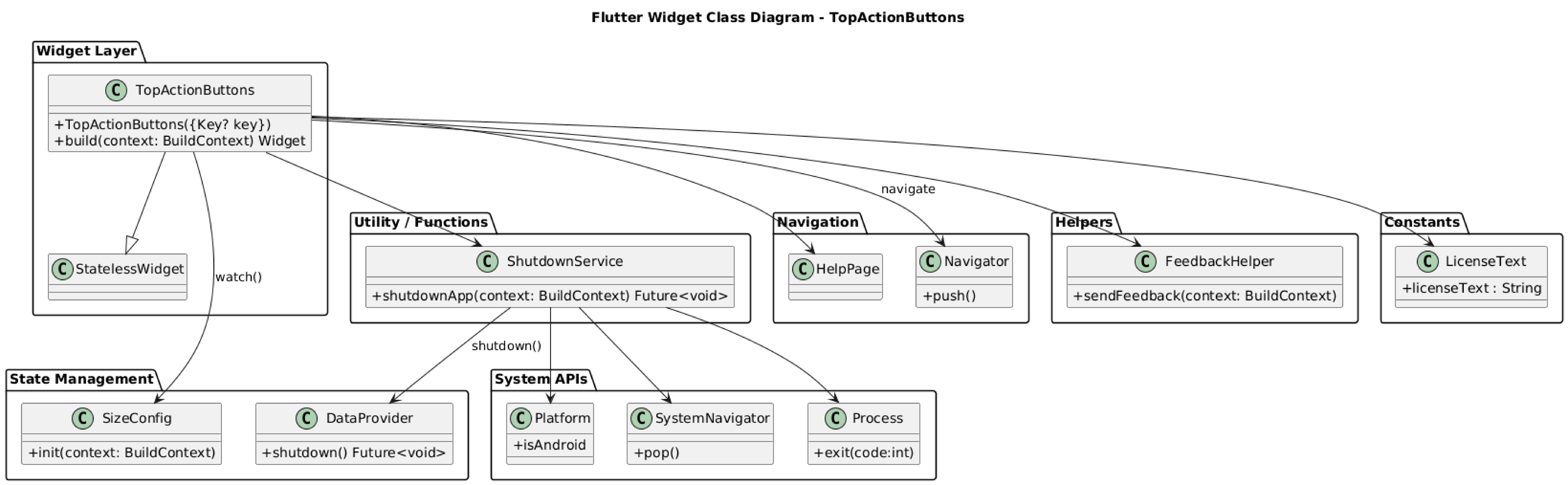

Flutter Widget Class Diagram - SetResetButtons
Core Types
BuildContext
Widget
List
bool
Widget Layer
SetResetButtons
-isVertical : bool
-selectedList : List<bool>
-maxWidth : double
-onUpdate : ValueChanged<List<bool>>
+SetResetButtons({
Key? key,
bool isVertical,
List<bool> selectedList,
double maxWidth,
ValueChanged<List<bool>> onUpdate
})
+build(context: BuildContext) : Widget
StatelessWidget
Flutter UI Framework
ToggleButtons
Axis
BorderRadius
Radius
Colors
BoxConstraints
callback
children
Callbacks
ValueChanged
T
+call(T value)
Application Resources
setresetlist
+List<Widget>


Flutter Widget Class Diagram - SudokuBlock
Flutter Core Types
BuildContext
Widget
List
Widget Layer
SudokuBlock
-block_id : int
+SudokuBlock({Key? key, required int block_id})
+build(context: BuildContext) : Widget
StatelessWidget
creates (9 elements)
index calculation
Application Widgets
SudokuElement
+element_id : int
Sudoku Constants
constSudokuNumRow
+value : int
Flutter UI Framework
Scaffold
Container
GridView
BoxDecoration
Border
Colors
NeverScrollableScrollPhysics
EdgeInsets

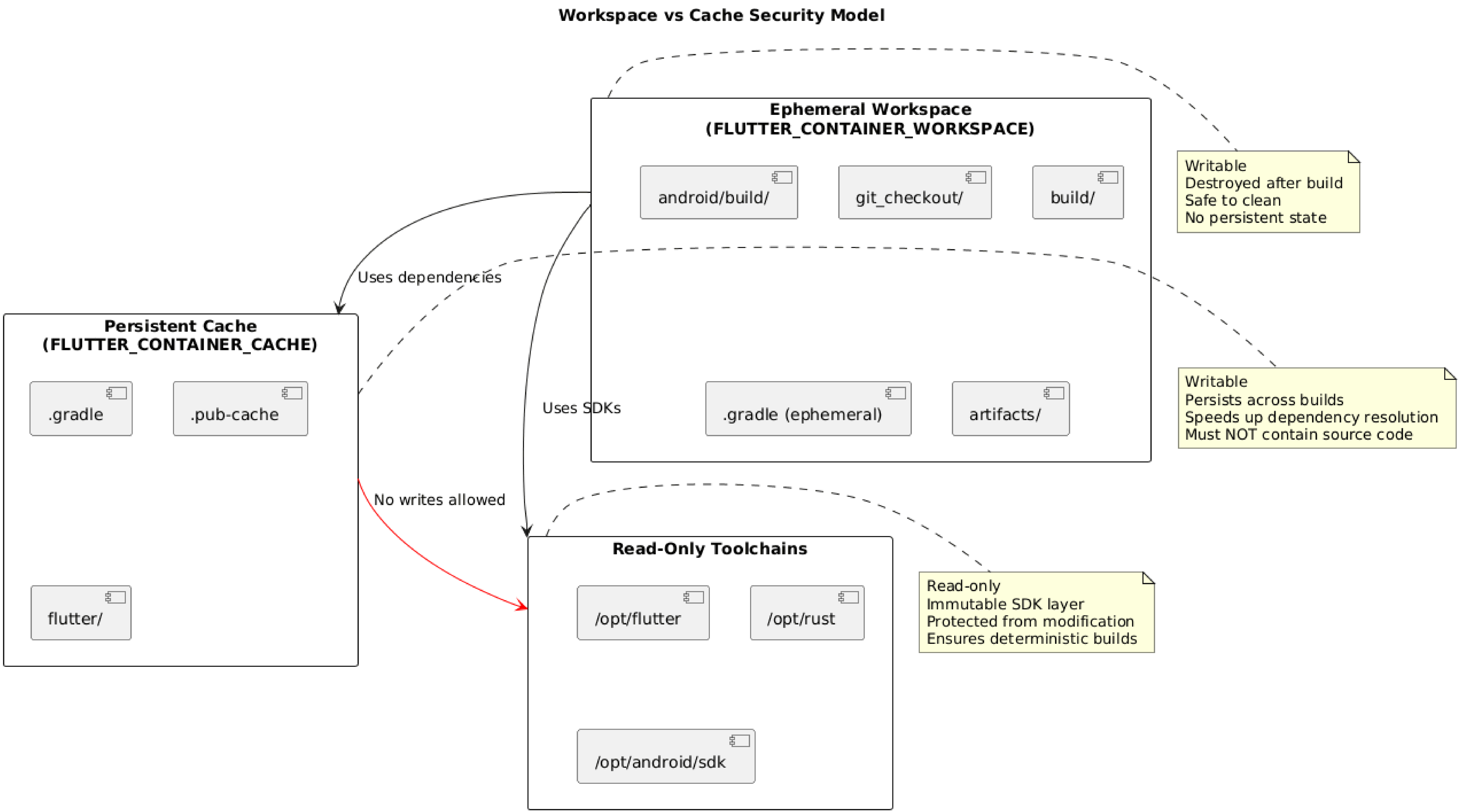
Flutter Widget Class Diagram - SudokuElement


Flutter Widget Class Diagram - SudokuGrid
Widget Layer
SudokuGrid
+SudokuGrid({Key? key})
+build(context: BuildContext) : Widget
StatelessWidget
watch()
creates 9 blocks
State Management
SizeConfig
+orientation
+screenWidth
+init(context: BuildContext)
Application Widgets
SudokuBlock
+block_id : int
Flutter Core Types
List
BuildContext
Flutter UI Framework
Widget
Scaffold
GridView
EdgeInsets

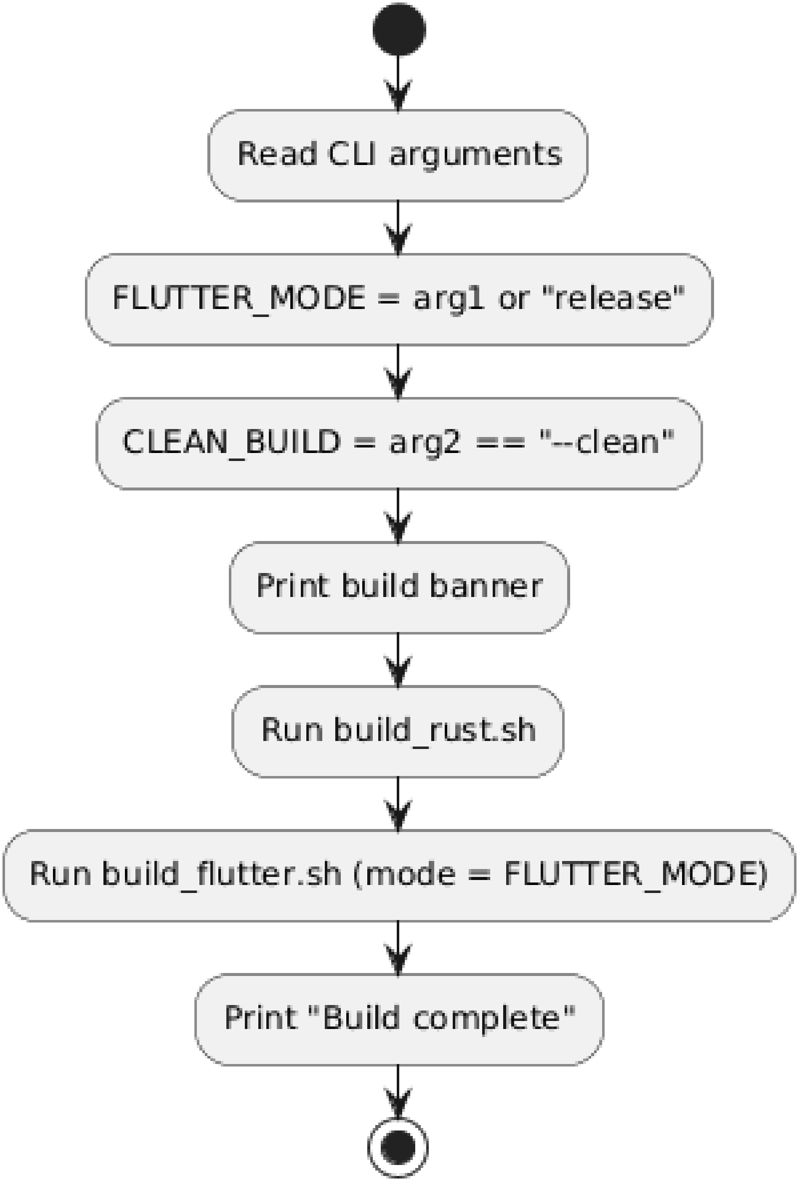
Flutter Widget Class Diagram - ToggleButtonsSample
Widget Layer
ToggleButtonsSample
+ToggleButtonsSample({Key? key})
+createState() : State<ToggleButtonsSample>
StatefulWidget
creates
Widget State
_ToggleButtonsSampleState
-_selectedNumberList : SelectedNumberList
-_selectedSetResetList : SelectedSetResetList
-_selectedPatternList : SelectedPatternList
-_vertical : bool
+build(context: BuildContext) : Widget
State
T
Dart Core
max()
List
bool
Flutter Core Types
Widget
BuildContext
watch()
State Management
DataProvider
+updateDataselectedSetResetList(List<bool>)
+updateDataNumberlist(List<bool>)
+updateDataselectedPatternList(List<bool>)
SizeConfig
+safeBlockBottomHMIGridVertical : double
+safeBlockBottomHMIGridHorizontal : double
+init(context: BuildContext)
Application Widgets
SetResetButtons
NumberButtons
PatternButtons
Flutter UI Framework
Scaffold
Center
Column
SizedBox
SingleChildScrollView


Flutter Widget Class Diagram - TopActionButtons
Widget Layer
TopActionButtons
+TopActionButtons({Key? key})
+build(context: BuildContext) Widget
StatelessWidget
watch()
Utility / Functions
ShutdownService
+shutdownApp(context: BuildContext) Future<void>
Navigation
HelpPage
Navigator
+push()
navigate
Helpers
FeedbackHelper
+sendFeedback(context: BuildContext)
Constants
LicenseText
+licenseText : String
shutdown()
State Management
SizeConfig
+init(context: BuildContext)
DataProvider
+shutdown() Future<void>
System APIs
Platform
+isAndroid
SystemNavigator
+pop()
Process
+exit(code:int)

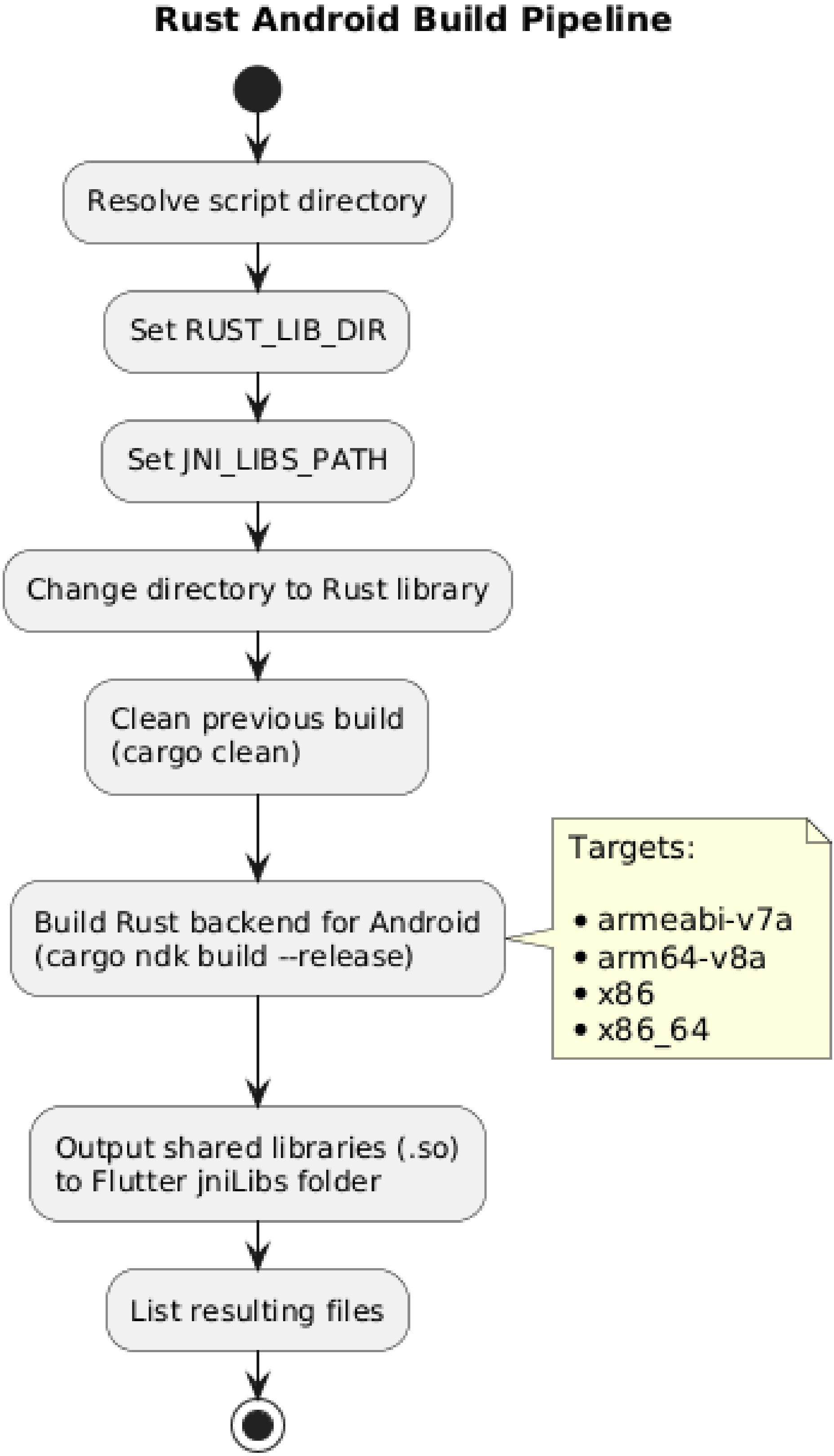
Flutter Widget Class Diagram - UndoIconButtons

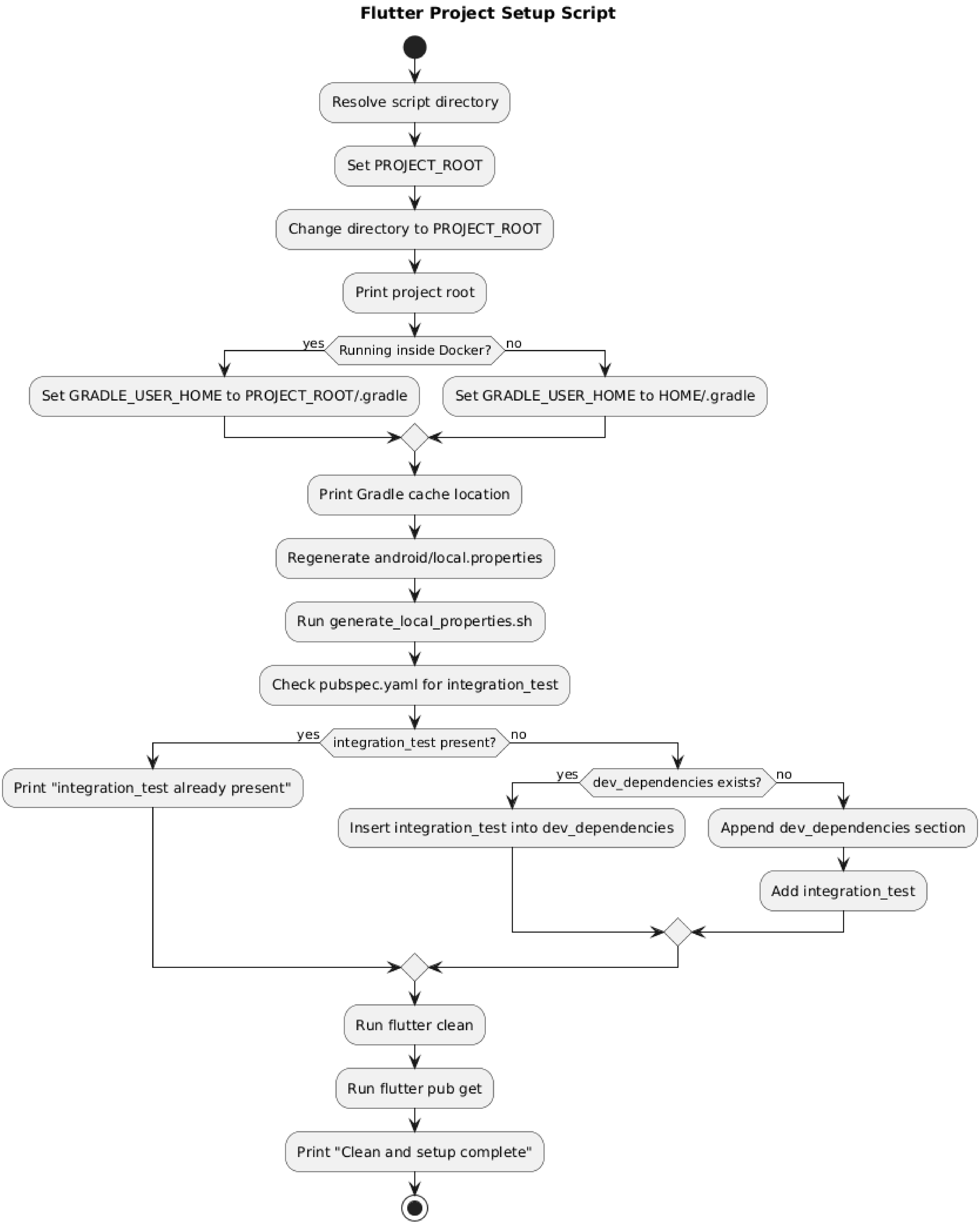

Workspace vs Cache Security Model
Ephemeral Workspace
(FLUTTER_CONTAINER_WORKSPACE)
android/build/
git_checkout/
build/
.gradle (ephemeral)
artifacts/
Writable
Destroyed after build
Safe to clean
No persistent state
Uses dependencies
Uses SDKs
Persistent Cache
(FLUTTER_CONTAINER_CACHE)
.gradle
.pub-cache
flutter/
Writable
Persists across builds
Speeds up dependency resolution
Must NOT contain source code
No writes allowed
Read-Only Toolchains
/opt/flutter
/opt/rust
/opt/android/sdk
Read-only
Immutable SDK layer
Protected from modification
Ensures deterministic builds

Rust + Flutter Build Pipeline

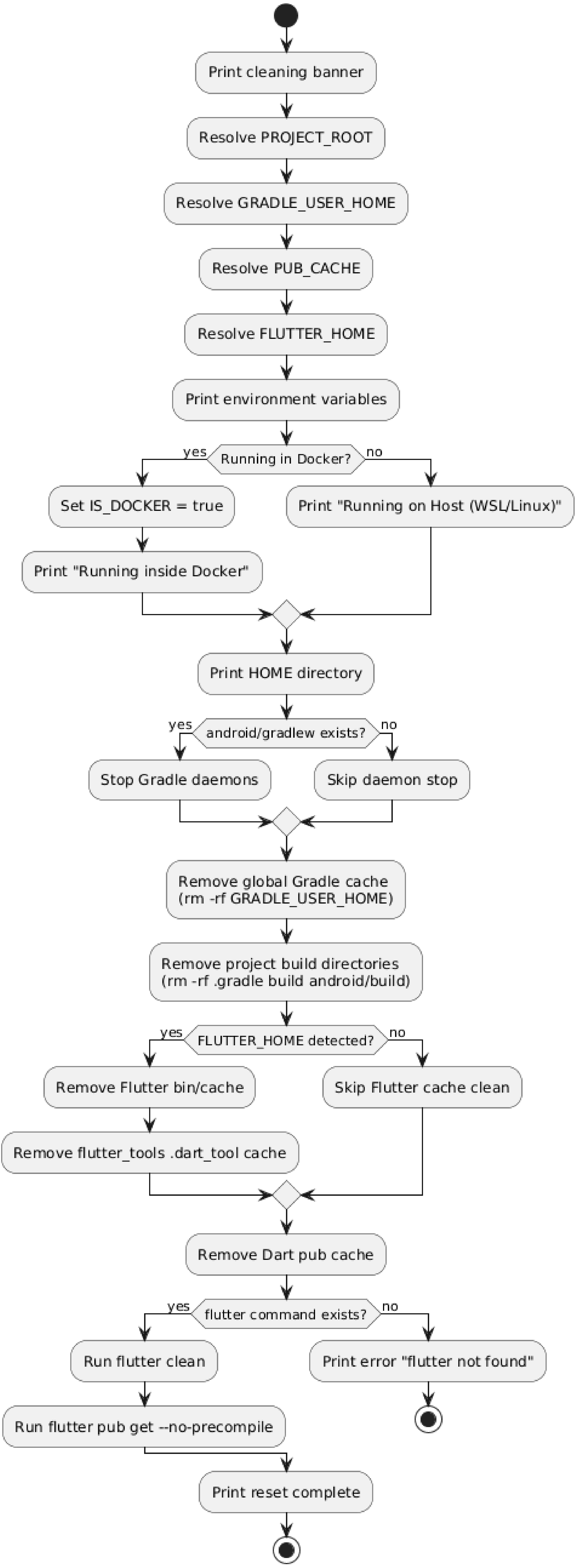


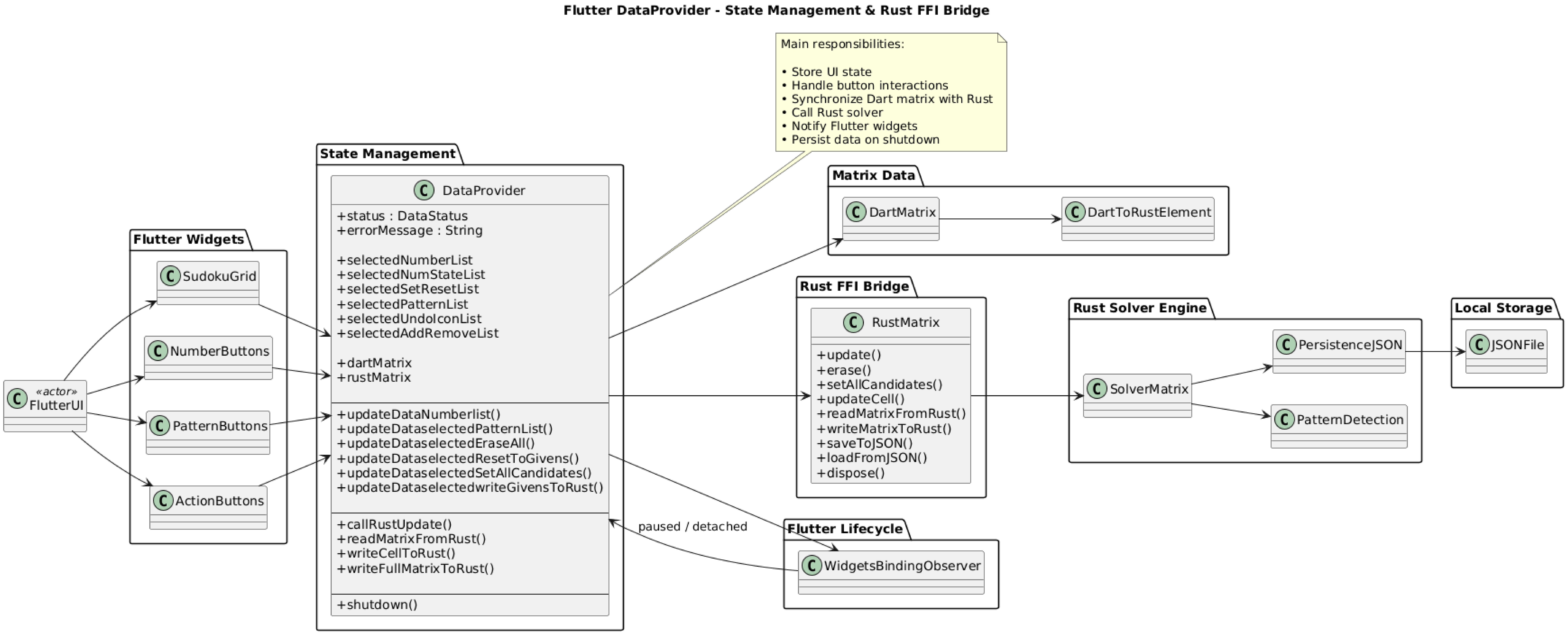
Flutter + Rust Build & Deploy Script

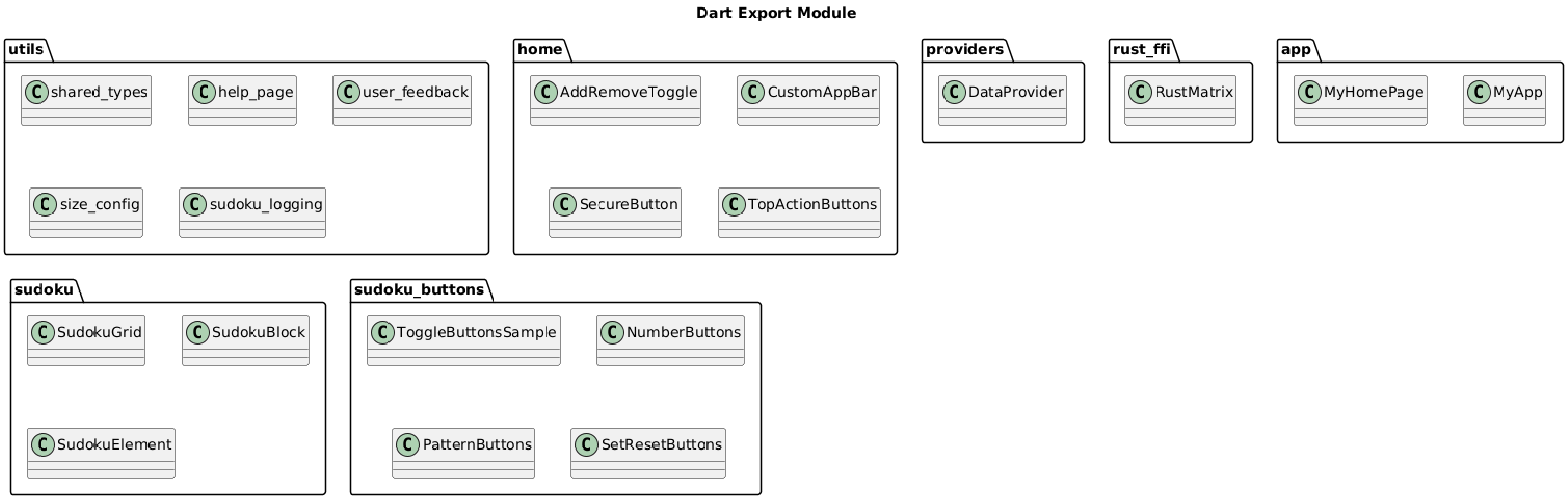

Rust Android Build Pipeline
Resolve script directory
Set RUST_LIB_DIR
Set JNI_LIBS_PATH
Change directory to Rust library
Clean previous build
(cargo clean)
Build Rust backend for Android
(cargo ndk build --release)
Targets:
• armeabi-v7a
• arm64-v8a
• x86
• x86_64
Output shared libraries (.so)
to Flutter jniLibs folder
List resulting files


Flutter Project Setup Script
Resolve script directory
Set PROJECT_ROOT
Change directory to PROJECT_ROOT
Print project root
Running inside Docker?
yes
no
Set GRADLE_USER_HOME to PROJECT_ROOT/.gradle
Set GRADLE_USER_HOME to HOME/.gradle
Print Gradle cache location
Regenerate android/local.properties
Run generate_local_properties.sh
Check pubspec.yaml for integration_test
integration_test present?
yes
no
Print "integration_test already present"
dev_dependencies exists?
yes
no
Insert integration_test into dev_dependencies
Append dev_dependencies section
Add integration_test
Run flutter clean
Run flutter pub get
Print "Clean and setup complete"

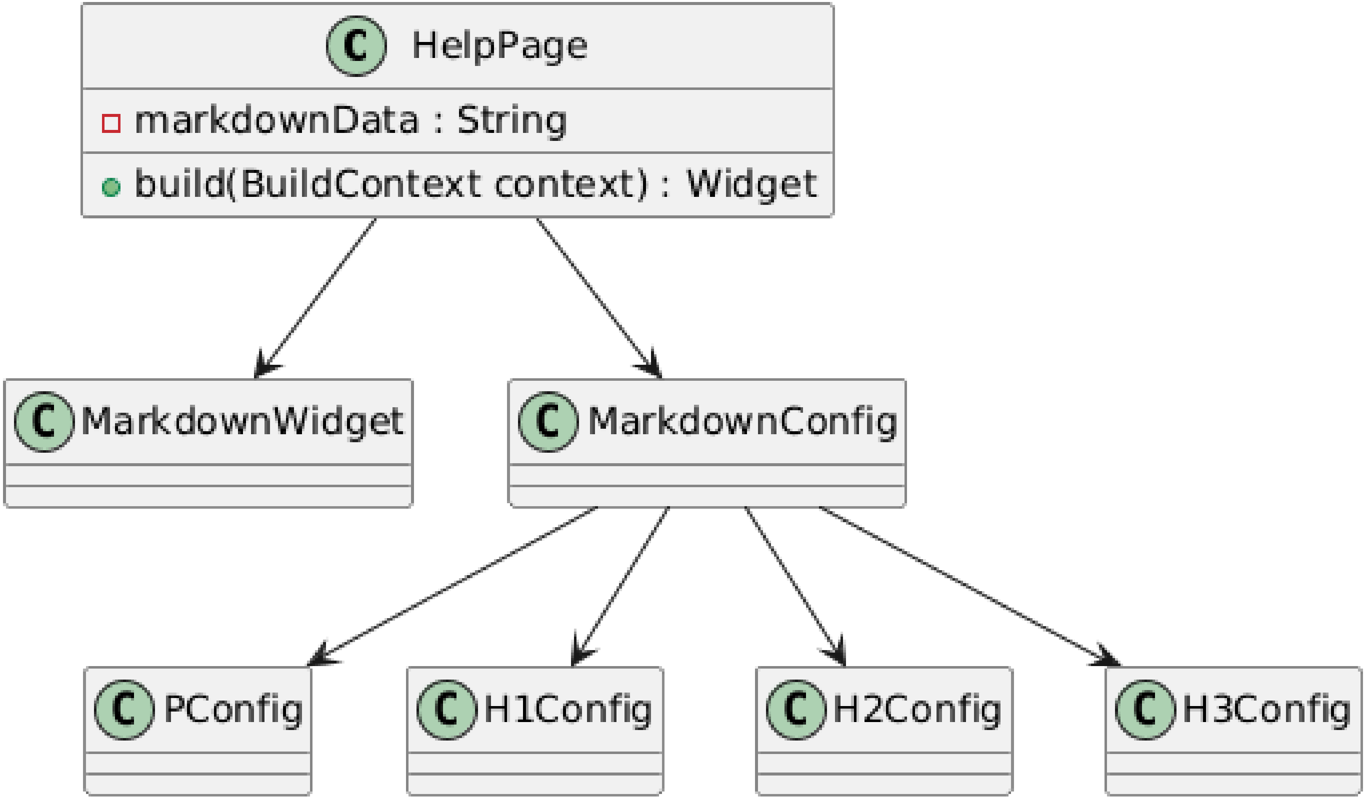
Flutter & Gradle Cache Reset Script

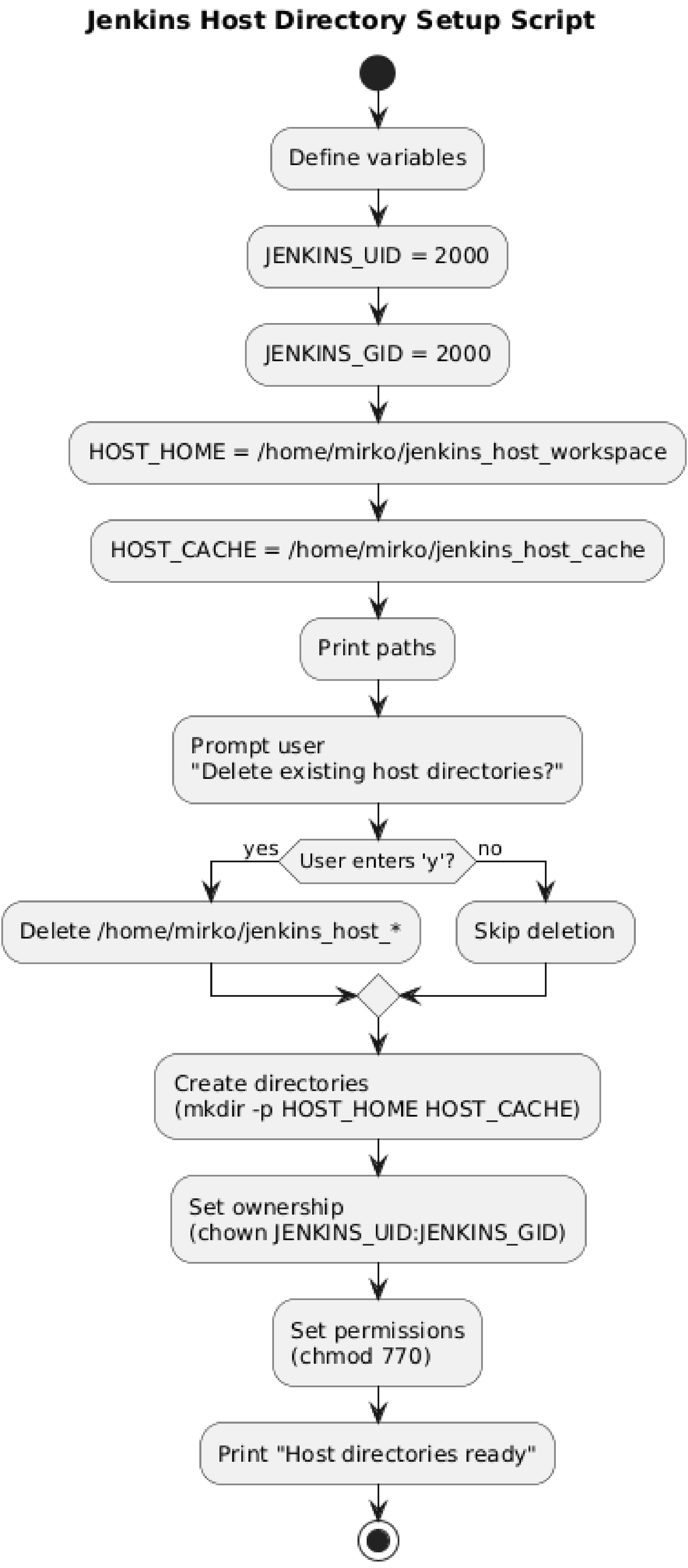


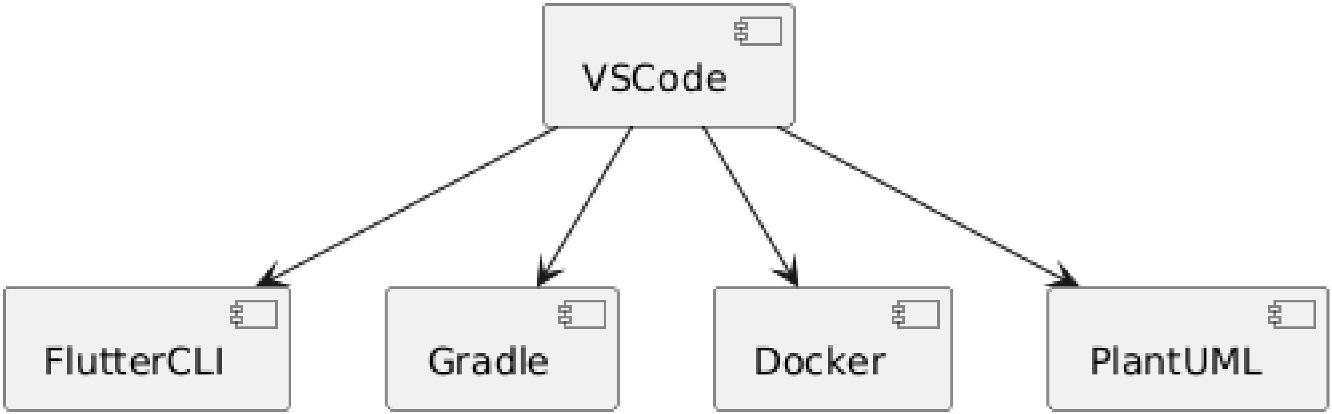

Flutter DataProvider - State Management & Rust FFI Bridge
Main responsibilities:
• Store UI state
• Handle button interactions
• Synchronize Dart matrix with Rust
• Call Rust solver
• Notify Flutter widgets
• Persist data on shutdown
State Management
DataProvider
+status : DataStatus
+errorMessage : String
+selectedNumberList
+selectedNumStateList
+selectedSetResetList
+selectedPatternList
+selectedUndoIconList
+selectedAddRemoveList
+dartMatrix
+rustMatrix
+updateDataNumberlist()
+updateDataselectedPatternList()
+updateDataselectedEraseAll()
+updateDataselectedResetToGivens()
+updateDataselectedSetAllCandidates()
+updateDataselectedwriteGivensToRust()
+callRustUpdate()
+readMatrixFromRust()
+writeCellToRust()
+writeFullMatrixToRust()
+shutdown()
Flutter Widgets
SudokuGrid
NumberButtons
PatternButtons
ActionButtons
«actor»
FlutterUI
Matrix Data
DartMatrix
DartToRustElement
Rust FFI Bridge
RustMatrix
+update()
+erase()
+setAllCandidates()
+updateCell()
+readMatrixFromRust()
+writeMatrixToRust()
+saveToJSON()
+loadFromJSON()
+dispose()
Rust Solver Engine
SolverMatrix
PersistenceJSON
PatternDetection
Local Storage
JSONFile
Flutter Lifecycle
WidgetsBindingObserver
paused / detached

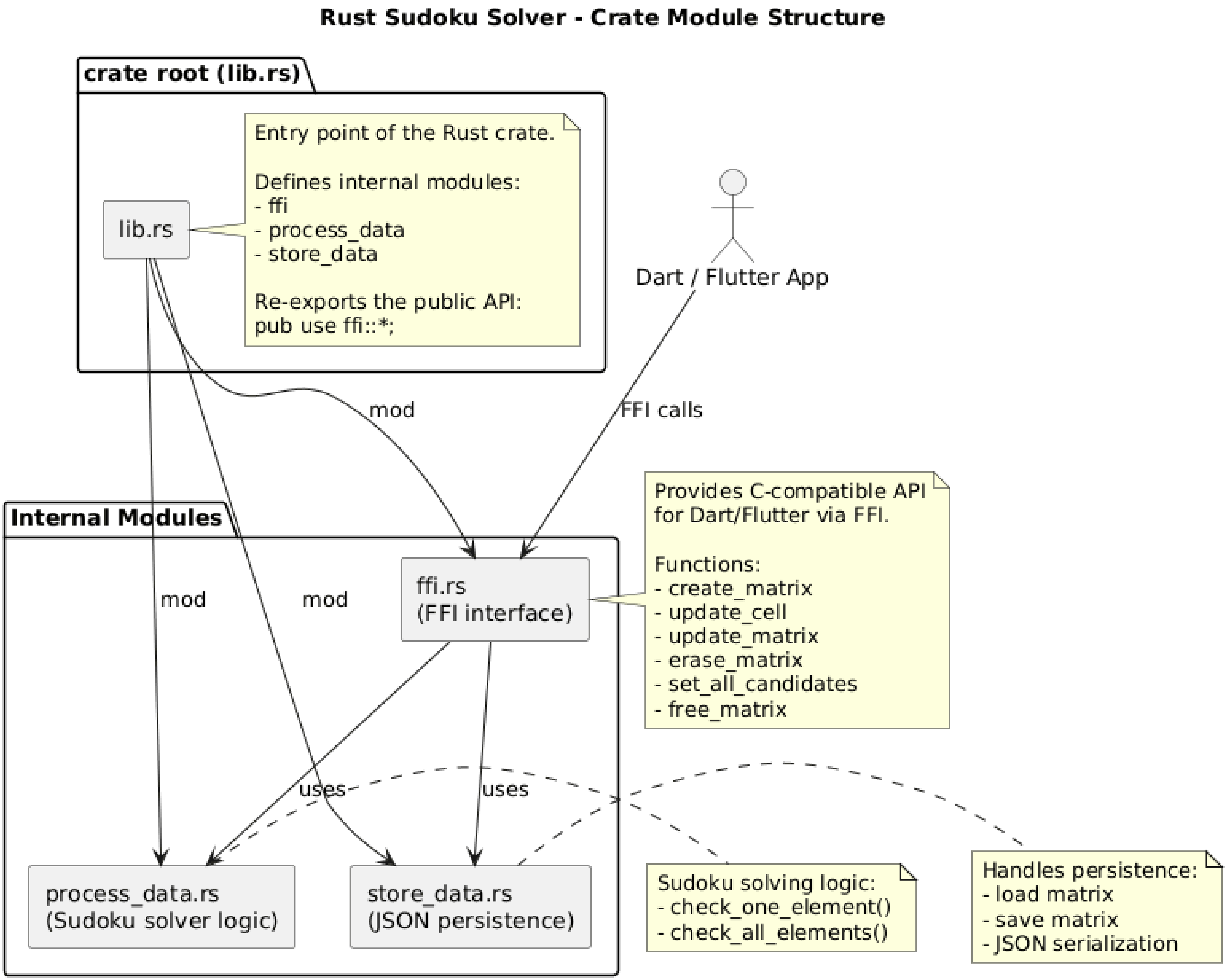

Dart Export Module
utils
shared_types
help_page
user_feedback
size_config
sudoku_logging
home
AddRemoveToggle
CustomAppBar
SecureButton
TopActionButtons
providers
DataProvider
rust_ffi
RustMatrix
app
MyHomePage
MyApp
sudoku
SudokuGrid
SudokuBlock
SudokuElement
sudoku_buttons
ToggleButtonsSample
NumberButtons
PatternButtons
SetResetButtons

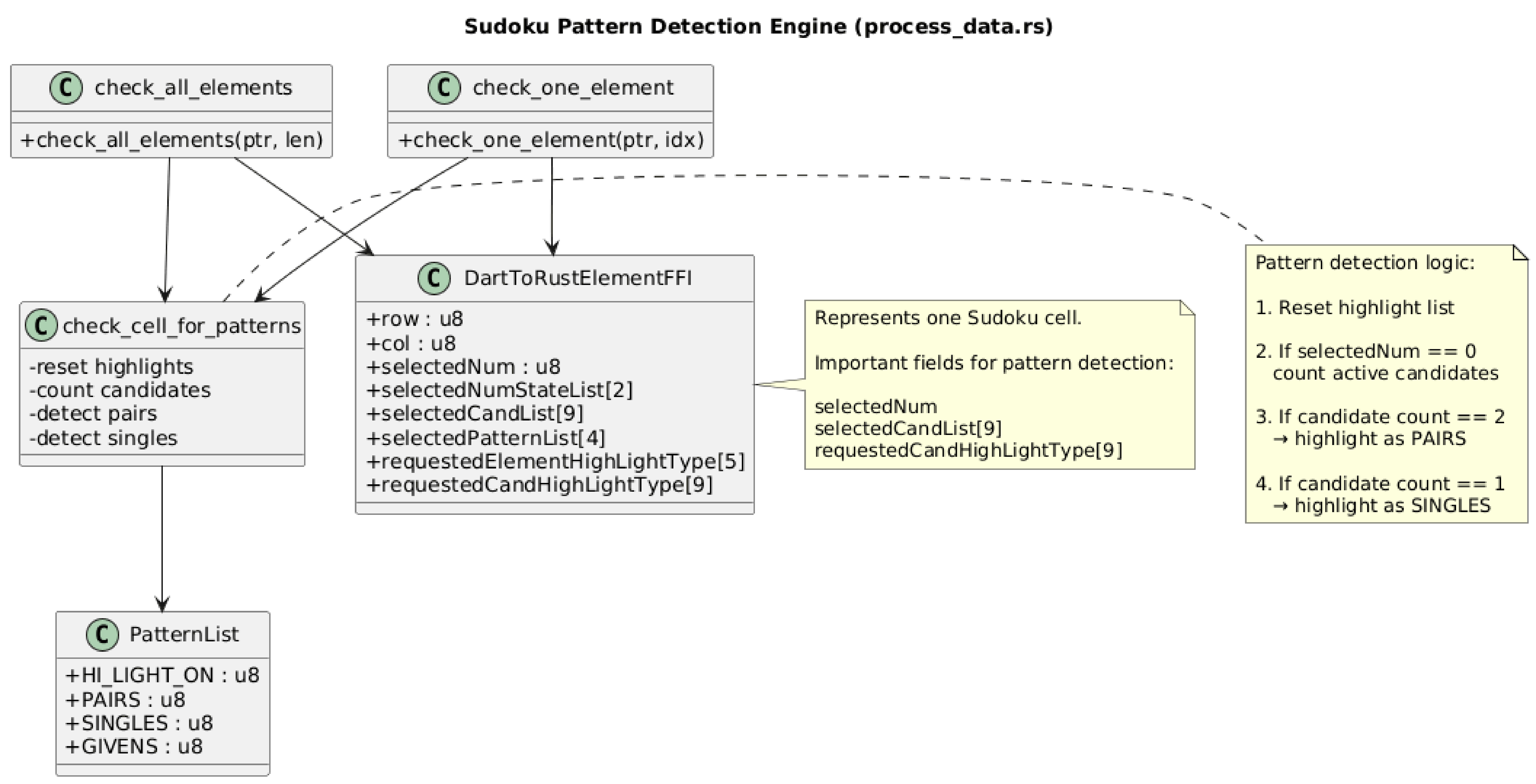
Rust Sudoku Matrix FFI Architecture
C FFI Interface
free_matrix
+free_matrix(ptr, rows, cols)
create_matrix
+create_matrix(rows, cols)
update_cell
+update_cell(ptr, rows, cols, idx)
update_matrix
+update_matrix(ptr, rows, cols)
erase_matrix
+erase_matrix(ptr, rows, cols, erase_givens)
set_all_candidates
+set_all_candidates(ptr, rows, cols)
Memory Model
MatrixMemory
+initialize cells
+contiguous memory block
+allocate()
Allocator
+alloc(Layout)
+dealloc(Layout)
contains 81
Pattern Detection Engine
check_one_element
check_all_elements
Detects patterns such as:
- pairs
- singles
- highlight candidates
Runs solver analysis
over entire matrix
FFI Data Model
DartToRustElementFFI
+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList[2]
+selectedCandList[9]
+selectedPatternList[4]
+requestedElementHighLightType[5]
+requestedCandHighLightType[9]
Represents a single Sudoku cell.
Used for FFI communication
between Dart and Rust.
Total cells = 81
Matrix size = 9 x 9
Constants
MatrixConstants
+MAX_UINT : u8 = 255
+CONST_MATRIX_SIZE : u8 = 9
+CONST_MATRIX_ELEMENTS : u8 = 81
PatternList
+HI_LIGHT_ON : u8
+PAIRS : u8
+SINGLES : u8
+GIVENS : u8
NumStateListIndex
+GIVENS : u8
+FUTUREUSE : u8
Default Data Arrays
AllCandidates
+constSelectedNumberListAllSelected[9]
DefaultLists
+constSelectedNumberList[9]
+constSelectedNumStateList[2]
+constSelectedPatternList[4]
+constRequestedElementHighLightType[5]
+constRequestedCandHighLightType[9]


HelpPage
markdownData : String
build(BuildContext context) : Widget
MarkdownWidget
MarkdownConfig
PConfig
H1Config
H2Config
H3Config


Jenkins Host Directory Setup Script
Define variables
JENKINS_UID = 2000
JENKINS_GID = 2000
HOST_HOME = /home/mirko/jenkins_host_workspace
HOST_CACHE = /home/mirko/jenkins_host_cache
Print paths
Prompt user
"Delete existing host directories?"
User enters 'y'?
yes
no
Delete /home/mirko/jenkins_host_*
Skip deletion
Create directories
(mkdir -p HOST_HOME HOST_CACHE)
Set ownership
(chown JENKINS_UID:JENKINS_GID)
Set permissions
(chmod 770)
Print "Host directories ready"

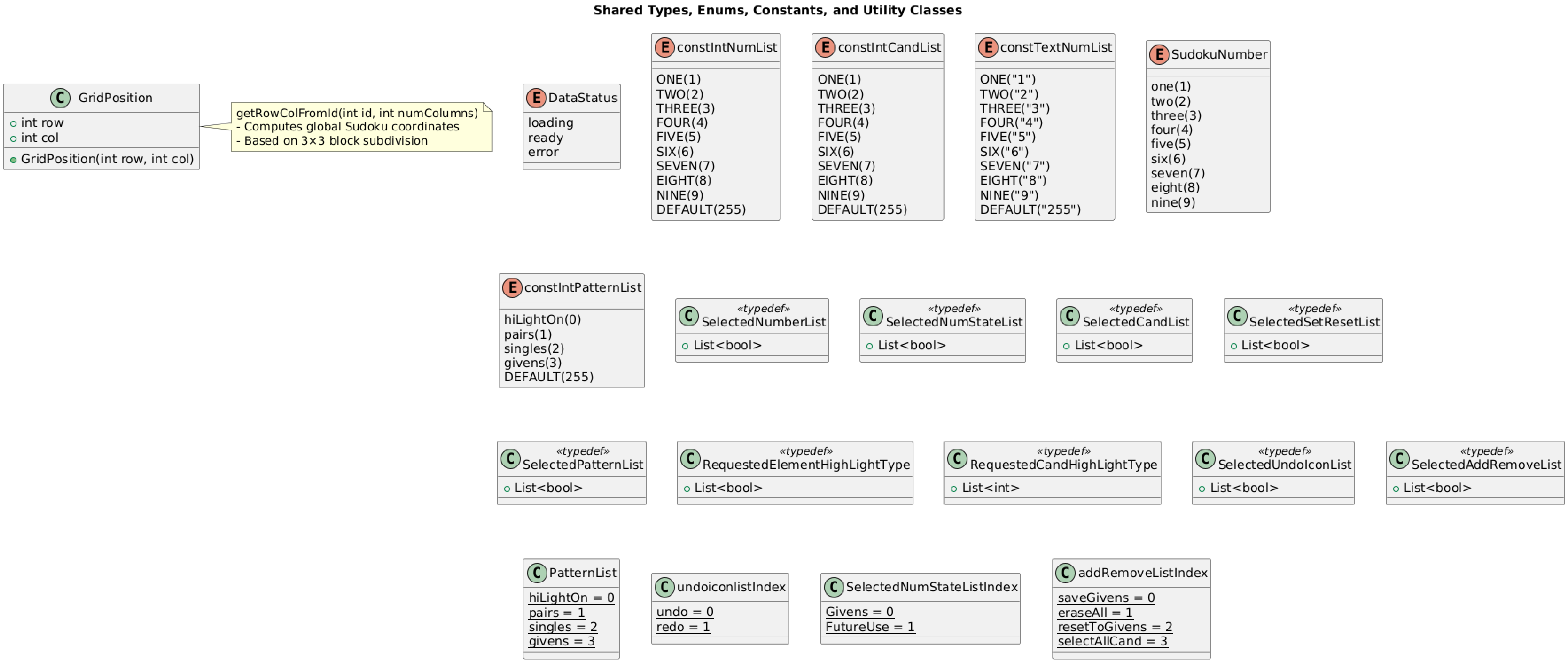

VSCode
FlutterCLI
Gradle
Docker
PlantUML


Rust Sudoku Solver - Crate Module Structure
crate root (lib.rs)
lib.rs
Entry point of the Rust crate.
Defines internal modules:
- ffi
- process_data
- store_data
Re-exports the public API:
pub use ffi::*;
Dart / Flutter App
mod
FFI calls
Internal Modules
ffi.rs
(FFI interface)
Provides C-compatible API
for Dart/Flutter via FFI.
Functions:
- create_matrix
- update_cell
- update_matrix
- erase_matrix
- set_all_candidates
- free_matrix
mod
mod
uses
uses
process_data.rs
(Sudoku solver logic)
store_data.rs
(JSON persistence)
Sudoku solving logic:
- check_one_element()
- check_all_elements()
Handles persistence:
- load matrix
- save matrix
- JSON serialization

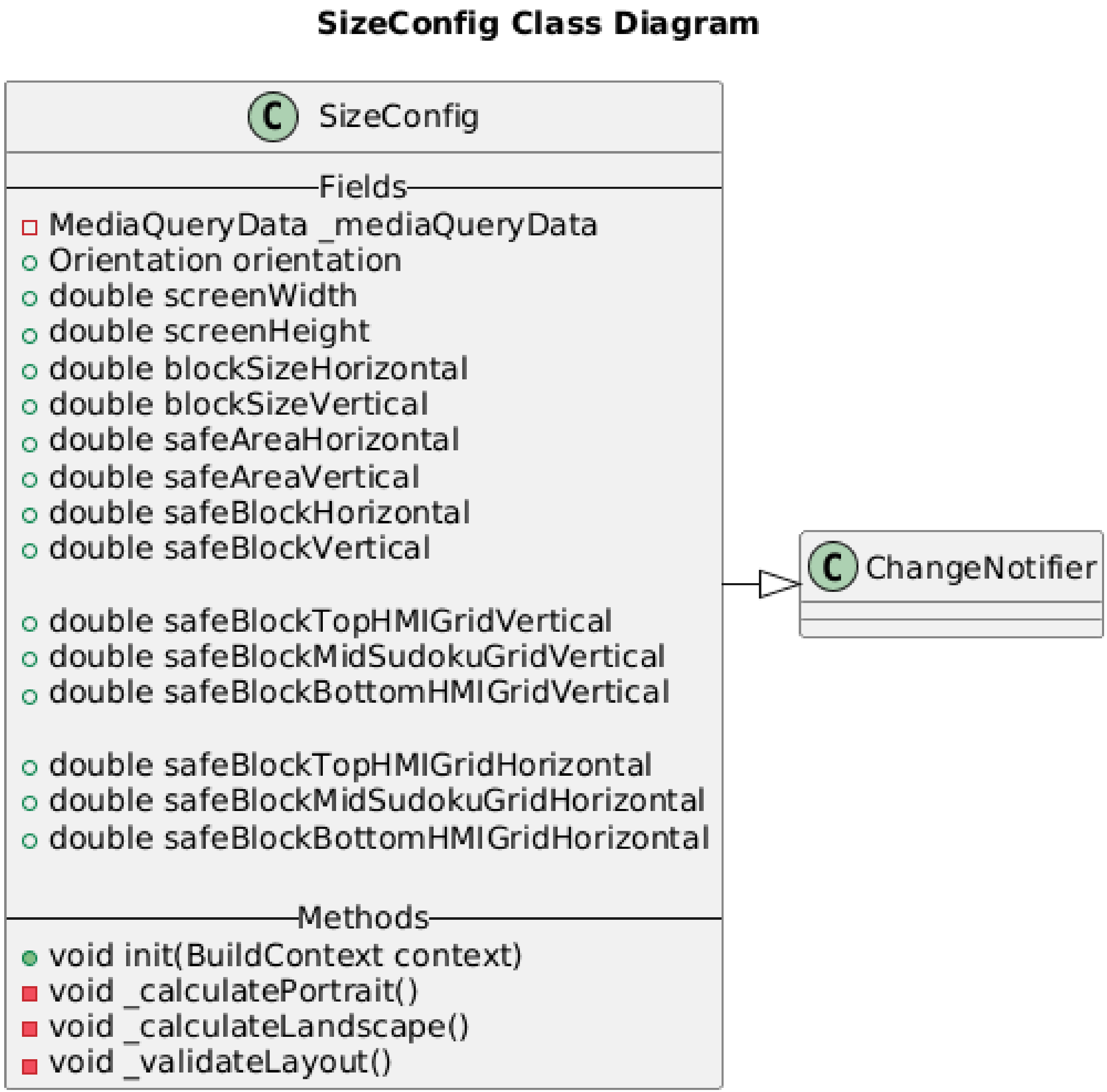

Sudoku Pattern Detection Engine (process_data.rs)
check_all_elements
+check_all_elements(ptr, len)
check_one_element
+check_one_element(ptr, idx)
check_cell_for_patterns
-reset highlights
-count candidates
-detect pairs
-detect singles
DartToRustElementFFI
+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList[2]
+selectedCandList[9]
+selectedPatternList[4]
+requestedElementHighLightType[5]
+requestedCandHighLightType[9]
Represents one Sudoku cell.
Important fields for pattern detection:
selectedNum
selectedCandList[9]
requestedCandHighLightType[9]
Pattern detection logic:
1. Reset highlight list
2. If selectedNum == 0
count active candidates
3. If candidate count == 2
→ highlight as PAIRS
4. If candidate count == 1
→ highlight as SINGLES
PatternList
+HI_LIGHT_ON : u8
+PAIRS : u8
+SINGLES : u8
+GIVENS : u8

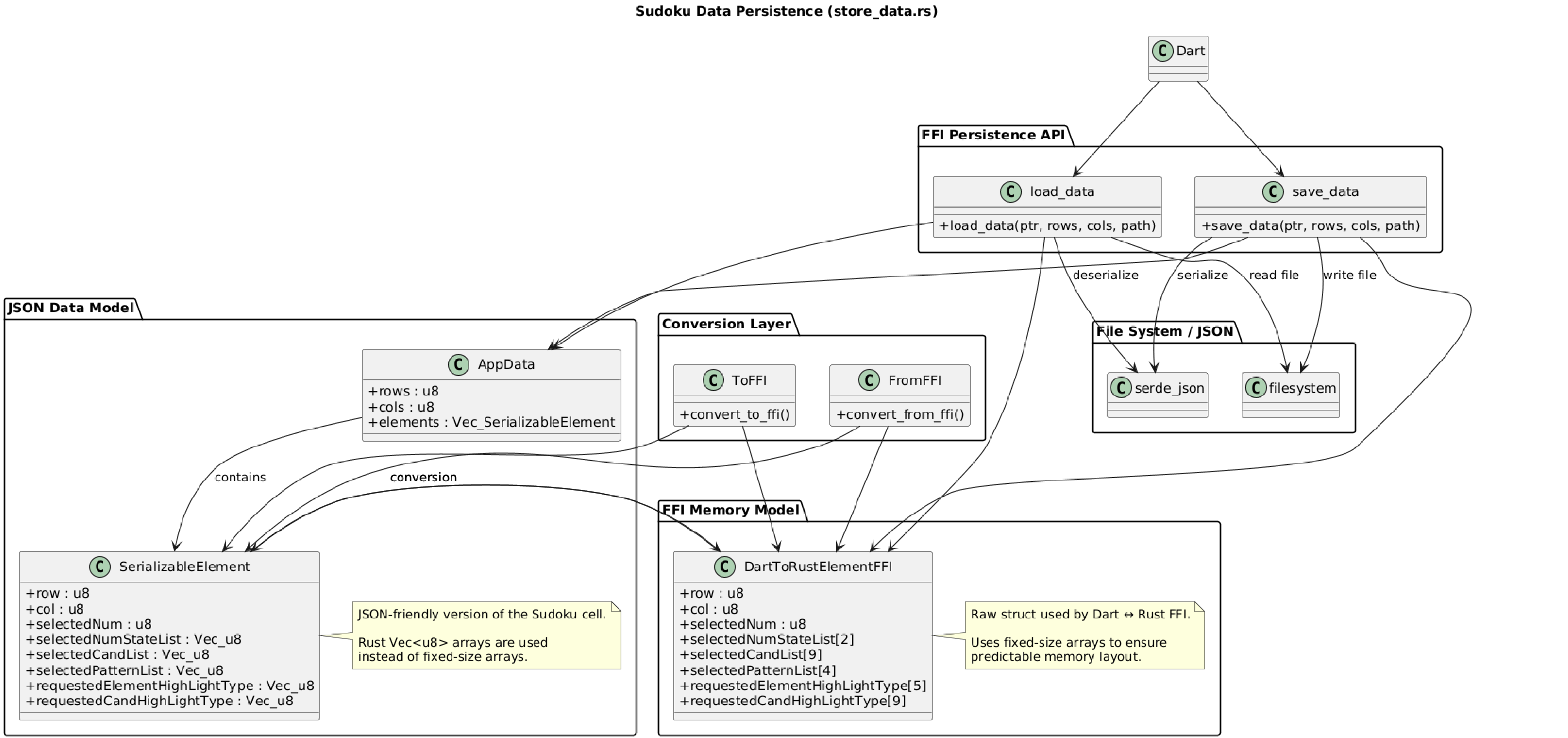
Dart ↔ Rust FFI Bridge (rust_matrix.dart)

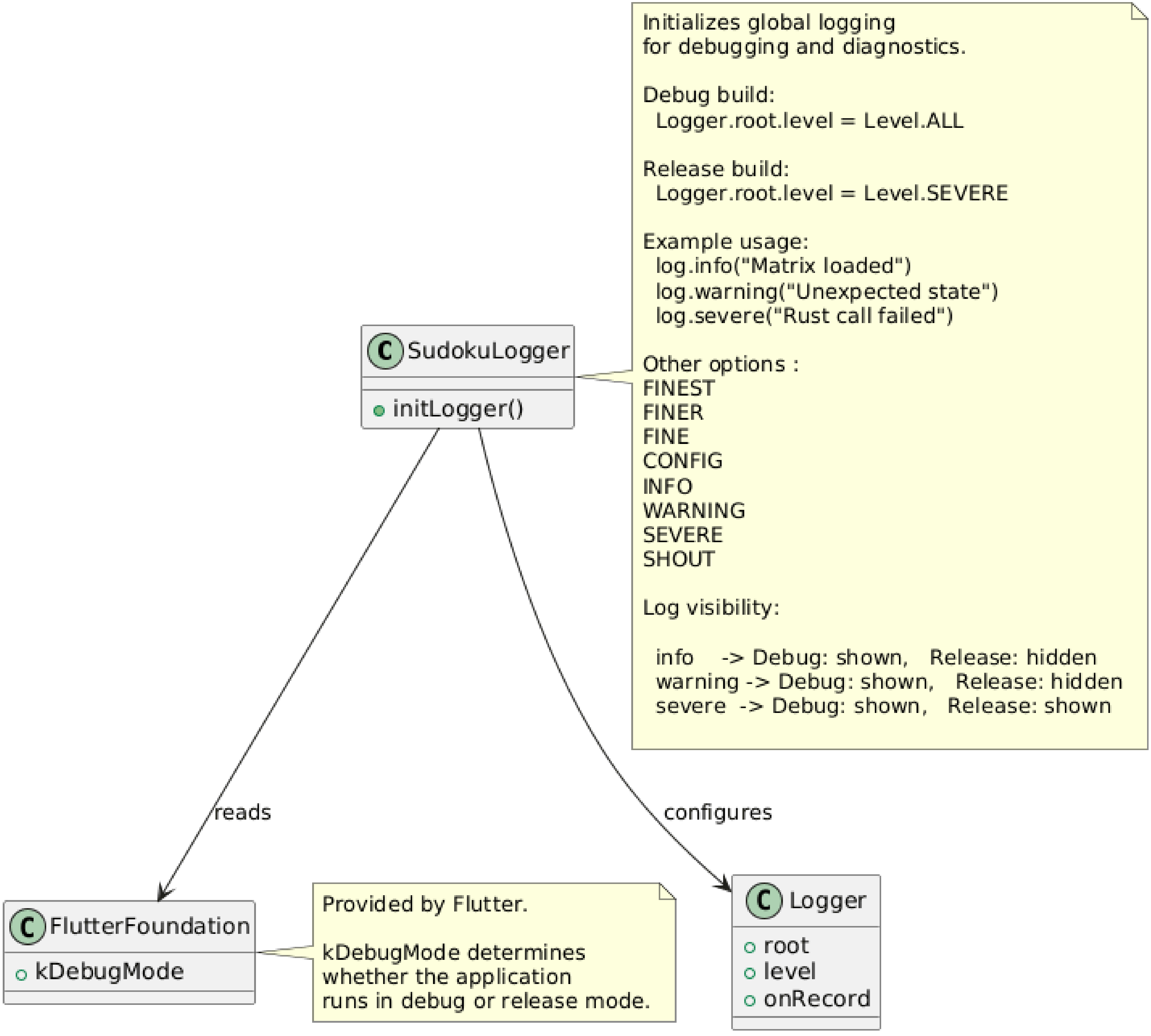

WSL ADB over TCP/IP Setup Workflow
Detect Windows host IP from /etc/resolv.conf
Host IP detected?
yes
no
Set WIN_IP
Print error
Configure ADB_SERVER_SOCKET
Running in WSL?
yes
no
Use Windows adb.exe
Use native adb
Persist ADB_SERVER_SOCKET in /.bashrc
Verify adb command exists
adb installed?
yes
no
Print install instructions
Check if Windows adb.exe process running
adb.exe running?
yes
no
Continue
Start adb.exe via PowerShell
Check ADB port binding (5037)
Bound to 127.0.0.1?
ok
no
Binding OK
Bound to 0.0.0.0?
yes
no
Restart adb.exe with correct binding
Warn user
Verify connectivity to Windows ADB server
Kill any WSL adb server
Verify connection with adb devices
Detect USB device
USB device found?
no
yes
Enable TCP/IP mode
Determine device IP
Saved IP exists?
yes
no
Load IP from file
Detect IP from device
Save IP to file
Ping device for connectivity
Attempt TCP/IP connection loop
adb connect DEVICE_IP:5555
Check adb devices list
device not connected
Connected?
yes
no
TCP/IP ADB ready
Print troubleshooting instructions
Display device list
Print user instructions
Detect WSL CMake
CMake installed?
yes
no
Export ANDROID_CMAKE
Print install instructions
Persist in /.bashrc
Configure Gradle CMake path
Persist DEVICE_IP in /.bashrc
Print setup complete



Shared Types, Enums, Constants, and Utility Classes
GridPosition
+ int row
+ int col
+ GridPosition(int row, int col)
getRowColFromId(int id, int numColumns)
- Computes global Sudoku coordinates
- Based on 3×3 block subdivision
DataStatus
loading
ready
error
constIntNumList
ONE(1) TWO(2) THREE(3) FOUR(4) FIVE(5) SIX(6) SEVEN(7) EIGHT(8) NINE(9) DEFAULT(255)
constIntCandList
ONE(1) TWO(2) THREE(3) FOUR(4) FIVE(5) SIX(6) SEVEN(7) EIGHT(8) NINE(9) DEFAULT(255)
constTextNumList
ONE("1") TWO("2") THREE("3") FOUR("4") FIVE("5") SIX("6") SEVEN("7") EIGHT("8") NINE("9") DEFAULT("255")
SudokuNumber
one(1) two(2) three(3) four(4) five(5) six(6) seven(7) eight(8) nine(9)
constIntPatternList
hiLightOn(0) pairs(1) singles(2) givens(3) DEFAULT(255)
«typedef» SelectedNumberList
+ List<bool>
«typedef» SelectedNumStateList
+ List<bool>
«typedef» SelectedCandList
+ List<bool>
«typedef» SelectedSetResetList
+ List<bool>
«typedef» SelectedPatternList
+ List<bool>
«typedef» RequestedElementHighLightType
+ List<bool>
«typedef» RequestedCandHighLightType
+ List<int>
«typedef» SelectedUndoIconList
+ List<bool>
«typedef» SelectedAddRemoveList
+ List<bool>
PatternList
hiLightOn = 0
pairs = 1
singles = 2
givens = 3
undoiconlistIndex
undo = 0
redo = 1
SelectedNumStateListIndex
Givens = 0
FutureUse = 1
addRemoveListIndex
saveGivens = 0
eraseAll = 1
resetToGivens = 2
selectAllCand = 3

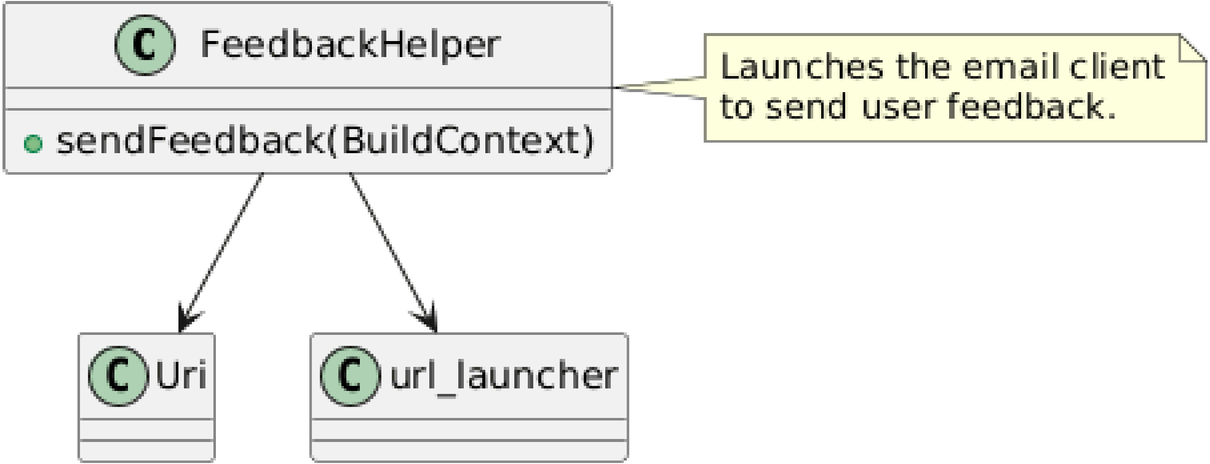

SizeConfig Class Diagram
SizeConfig
Fields
MediaQueryData _mediaQueryData
Orientation orientation
double screenWidth
double screenHeight
double blockSizeHorizontal
double blockSizeVertical
double safeAreaHorizontal
double safeAreaVertical
double safeBlockHorizontal
double safeBlockVertical
double safeBlockTopHMIGridVertical
double safeBlockMidSudokuGridVertical
double safeBlockBottomHMIGridVertical
double safeBlockTopHMIGridHorizontal
double safeBlockMidSudokuGridHorizontal
double safeBlockBottomHMIGridHorizontal
Methods
void init(BuildContext context)
void _calculatePortrait()
void _calculateLandscape()
void _validateLayout()
ChangeNotifier


Sudoku Data Persistence (store_data.rs)
Dart
FFI Persistence API
load_data
+load_data(ptr, rows, cols, path)
save_data
+save_data(ptr, rows, cols, path)
deserialize
serialize
read file
write file
JSON Data Model
AppData
+rows : u8
+cols : u8
+elements : Vec_SerializableElement
contains
conversion
Conversion Layer
ToFFI
+convert_to_ffi()
FromFFI
+convert_from_ffi()
File System / JSON
serde_json
filesystem
FFI Memory Model
SerializableElement
+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList : Vec_u8
+selectedCandList : Vec_u8
+selectedPatternList : Vec_u8
+requestedElementHighLightType : Vec_u8
+requestedCandHighLightType : Vec_u8
JSON-friendly version of the Sudoku cell.
Rust Vec<u8> arrays are used
instead of fixed-size arrays.
DartToRustElementFFI
+row : u8
+col : u8
+selectedNum : u8
+selectedNumStateList[2]
+selectedCandList[9]
+selectedPatternList[4]
+requestedElementHighLightType[5]
+requestedCandHighLightType[9]
Raw struct used by Dart ↔ Rust FFI.
Uses fixed-size arrays to ensure
predictable memory layout.



SudokuLogger
+ initLogger()
Initializes global logging
for debugging and diagnostics.
Debug build:
Logger.root.level = Level.ALL
Release build:
Logger.root.level = Level.SEVERE
Example usage:
log.info("Matrix loaded")
log.warning("Unexpected state")
log.severe("Rust call failed")
Other options :
FINEST
FINER
FINE
CONFIG
INFO
WARNING
SEVERE
SHOUT
Log visibility:
info -> Debug: shown, Release: hidden
warning -> Debug: shown, Release: hidden
severe -> Debug: shown, Release: shown
reads
configures
FlutterFoundation
+ kDebugMode
Provided by Flutter.
kDebugMode determines
whether the application
runs in debug or release mode.
Logger
+ root
+ level
+ onRecord

VSCode Tasks - Flutter + Rust Build Pipeline
Developer
VSCode Tasks
setup-adb-wsl
clean-flutter-docker
clean-flutter-host
clean-gradle-cache-host
clean-gradle-cache-docker
integration-test-usb-tcp-host
integration-test-ci-chrome-headless
Generate PlantUML PDF
build-all-host-release
build-all-docker-debug
build-all-docker-profile
build-all-docker-release
build-all-host-debug
build-all-host-profile
build-flutter-host-debug
build-flutter-docker-debug
Project Scripts
build_all.sh
build_flutter.sh
setup_adb_wsl.sh
clean_flutter.sh
clean_gradle_cache.sh
run_integration_test.sh
generate_plantuml_pdf.sh
Execution Environment
Docker Container
flutter_rust_env
Host (WSL/Linux)
Headless Chrome
Android Device


FeedbackHelper
sendFeedback(BuildContext)
Launches the email client
to send user feedback.
Uri
url_launcher